

CME 307 / MS&E 311 / OIT 676: Optimization

Operators

Professor Udell

Management Science and Engineering
Stanford

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Outline

Subgradients

Subgradient properties

Subgradient method

Proximal operators

Proximal gradient method

Relations

Fixed points

Averaged operators

Proximal method

Basic inequality

recall basic inequality for convex differentiable f :

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

- ▶ first-order approximation of f at x is global underestimator
- ▶ $(\nabla f(x), -1)$ supports **epi** f at $(x, f(x))$

what if f is not differentiable?

Non-differentiable functions

are these functions differentiable?

- ▶ $|t|$ for $t \in \mathbb{R}$
- ▶ $\|x\|_1$ for $x \in \mathbb{R}^n$
- ▶ $\|X\|_*$ for $X \in \mathbb{R}^{n \times n}$
- ▶ $\max_i a_i^T x + b_i$ for $x \in \mathbb{R}^n$
- ▶ $\lambda_{\max}(X)$ for $X \in \mathbb{R}^{n \times n}$
- ▶ indicators of convex sets $\mathcal{C}$

if not, where? can we find underestimators for them?

Subgradient of a function

g is a **subgradient** of f (not necessarily convex) at x if

$$f(y) \geq f(x) + g^T(y - x) \quad \text{for all } y$$

picture

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A: Yes, if f is nonsmooth at x

Q: Can a function f have no subgradient at a point x ?

A: Yes, if x does not lie on convex hull of f

Subgradients and convexity

- ▶ g is a subgradient of f at x iff $(g, -1)$ supports **epi** f at $(x, f(x))$
- ▶ g is a subgradient iff $f(x) + g^T(y - x)$ is a global (affine) underestimator of f
- ▶ if f is convex and differentiable, $\nabla f(x)$ is a subgradient of f at x

subgradients come up in several contexts:

- ▶ algorithms for nondifferentiable convex optimization
- ▶ convex analysis, e.g., optimality conditions, duality for nondifferentiable problems

(if $f(y) \leq f(x) + g^T(y - x)$ for all y , then g is a **supergradient**)

Subdifferential

set of all subgradients of f at x is called the **subdifferential** of f at x , denoted $\partial f(x)$

$$\partial f(x) = \{g : f(y) \geq f(x) + g^T(y - x) \quad \forall y\}$$

for any f ,

- ▶ $\partial f(x)$ is a closed convex set (can be empty)
- ▶ $\partial f(x) = \emptyset$ if $f(x) = \infty$

proof: use the definition

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if f is convex,

- ▶ $\partial f(x)$ is nonempty, for $x \in \mathbf{relint\,dom\,}f$
- ▶ $\partial f(x) = \{\nabla f(x)\}$, if f is differentiable at x
- ▶ if $\partial f(x) = \{g\}$, then f is differentiable at x and $g = \nabla f(x)$

Compute subgradient via definition

$g \in \partial f(x)$ iff

$$f(y) \geq f(x) + g^T(y - x) \quad \forall y \in \mathbf{dom}(f)$$

example. let $f(x) = |x|$ for $x \in \mathbb{R}$. suppose $s \in \mathbf{sign}(x)$, where

$$\mathbf{sign}(x) = \begin{cases} \{1\} & x > 0 \\ [-1, 1] & x = 0 \\ -\{1\} & x < 0. \end{cases}$$

then

$$f(y) = \max(y, -y) \geq sy = s(x + y - x) = |x| + s(y - x)$$

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so $\mathbf{sign}(x) \subseteq \partial f(x)$ (in fact, holds with equality)

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example. let $f(x) = \max_i a_i^T x + b_i$. then for any i ,

$$\begin{aligned} f(y) &= \max_i a_i^T y + b_i \\ &\geq a_i^T y + b_i \\ &= a_i^T (x + y - x) + b_i \\ &= a_i^T x + b_i + a_i^T (y - x) \\ &= f(x) + a_i^T (y - x), \end{aligned}$$

where the last line holds for $i \in \operatorname{argmax}_j a_j^T x + b_j$. so

- ▶ $a_i \in \partial f(x)$ for each $i \in \operatorname{argmax}_j a_j^T x + b_j$
- ▶ $\partial f(x)$ is convex, so

$$\text{Co}\{a_i : i \in \operatorname{argmax}_j a_j^T x + b_j\} \subseteq \partial f(x)$$

Compute subgradient via definition

$$g \in \partial f(x) \iff f(y) \geq f(x) + g^T(y - x) \quad \forall y \in \text{dom}(f)$$

example. let $f(X) = \lambda_{\max}(X)$.

Compute subgradient via definition

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example. let $f(X) = \lambda_{\max}(X)$. then

$$\begin{aligned} f(Y) &= \sup_{\|v\| \leq 1} v^T Y v \\ &= \sup_{\|v\| \leq 1} v^T (X + Y - X) v, \quad \|v\| \leq 1 \\ &= \sup_{\|v\| \leq 1} \left(v^T X v + v^T (Y - X) v \right), \quad \|v\| \leq 1 \\ &= v^T X v + \text{tr}(v v^T (Y - X)), \quad v \in \underset{\|v\| \leq 1}{\text{argmax}} v^T X v \\ &= \lambda_{\max}(X) + \text{tr}(v v^T (Y - X)), \quad v \in \underset{\|v\| \leq 1}{\text{argmax}} v^T X v \end{aligned}$$

- ▶ $v v^T \in \partial f(X)$ for each $v \in \underset{\|v\| \leq 1}{\text{argmax}} v^T X v$
- ▶ $\partial f(x)$ is convex, so

$$\text{Co}\{v v^T : v \in \underset{\|v\| \leq 1}{\text{argmax}} v^T X v\} \subseteq \partial f(x)$$

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Properties of subgradients

subgradient inequality:

$$g \in \partial f(x) \iff f(y) \geq f(x) + g^T(y - x) \quad \forall y \in \mathbf{dom}(f)$$

for convex f , we'll show

- ▶ subgradients are monotone: for any $x, y \in \mathbf{dom} f$, $g_y \in \partial f(y)$, and $g_x \in \partial f(x)$,

$$(g_y - g_x)^T(y - x) \geq 0$$

- ▶ $\partial f(x)$ is continuous: if f is (lower semi-)continuous, $x^{(k)} \rightarrow x$, $g^{(k)} \rightarrow g$, and $g^{(k)} \in \partial f(x^{(k)})$ for each k , then $g \in \partial f(x)$
- ▶ $\partial f(x) = \operatorname{argmax} g^T x - f(x)$

these will help us compute subgradients

Subgradients are monotone

fact. for any $x, y \in \text{dom } f$, $g_y \in \partial f(y)$, and $g_x \in \partial f(x)$,

$$(g_y - g_x)^T (y - x) \geq 0$$

proof. same as for differentiable case:

$$f(y) \geq f(x) + g_x^T (y - x) \quad f(x) \geq f(y) + g_y^T (x - y)$$

add these to get

$$(g_y - g_x)^T (y - x) \geq 0$$

Subgradients are preserved under limits

subgradient inequality:

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fact. if f is (lower semi-)continuous, $x^{(k)} \rightarrow x$, $g^{(k)} \rightarrow g$, and $g^{(k)} \in \partial f(x^{(k)})$ for each k , then $g \in \partial f(x)$

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proof. For each k and for every y ,

$$\begin{aligned} f(y) &\geq f(x^{(k)}) + (g^{(k)})^T(y - x^{(k)}) \\ \lim_{k \rightarrow \infty} f(y) &\geq \lim_{k \rightarrow \infty} f(x^{(k)}) + (g^{(k)})^T(y - x^{(k)}) \\ f(y) &\geq f(x) + g^T(y - x) \end{aligned}$$

moral. To find a subgradient $g \in \partial f(x)$, find points $x^{(k)} \rightarrow x$ where f is differentiable, and let $g = \lim_{k \rightarrow \infty} \nabla f(x^{(k)})$.

Subgradients are preserved under limits: example

consider $f(x) = |x|$. we know

$$\partial f(x) = \begin{cases} \{-1\} & x < 0 \\ ? & x = 0 \\ \{1\} & x > 0 \end{cases}$$

so

- ▶ $\lim_{x \rightarrow 0^+} \nabla(x) = 1$
- ▶ $\lim_{x \rightarrow 0^-} \nabla(x) = -1$

hence

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hence

- ▶ $-1 \in \partial f(0)$ and $1 \in \partial f(0)$
- ▶ $\partial f(0)$ is convex, so $[-1, 1] \subseteq \partial f(0)$
- ▶ and $\partial f(0)$ is monotone, so $[-1, 1] = \partial f(0)$

Convex functions can't be very non-differentiable

Theorem

Rockafellar, Convex Analysis, Thm 25.5 a convex function f is differentiable almost everywhere on the interior of its domain.

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corollary: pick $x \in \text{dom } f$ uniformly at random. then f is differentiable at x w/prob 1.

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Theorem

Rockafellar, Convex Analysis, Thm 25.5 a convex function f is differentiable almost everywhere on the interior of its domain.

corollary: pick $x \in \text{dom } f$ uniformly at random. then f is differentiable at x w/prob 1.

corollary: For a convex function f and any x , there is a sequence of points $x^{(k)} \rightarrow x$ where f is differentiable.

Subgradients and fenchel conjugates

fact. $g \in \partial f(x) \iff f^*(g) + f(x) = g^T x$

(recall the conjugate function $f^*(g) = \sup_x g^T x - f(x)$.)

Subgradients and fenchel conjugates

proof. if $f^*(g) + f(x) = g^T x$,

$$\begin{aligned} f^*(g) &= \sup_y g^T y - f(y) \\ &\geq g^T y - f(y) \quad \forall y \\ f(y) &\geq g^T y - f^*(g) \quad \forall y \\ &= g^T y - g^T x + f(x) \quad \forall y \\ &= g^T (y - x) + f(x) \quad \forall y \end{aligned}$$

so $g \in \partial f(x)$. conversely, if $g \in \partial f(x)$,

Subgradients and fenchel conjugates

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so $g \in \partial f(x)$. conversely, if $g \in \partial f(x)$,

$$\begin{aligned} f(y) &\geq g^T (y - x) + f(x) \\ g^T x - f(x) &\geq g^T y - f(y) \\ \sup_y g^T x - f(x) &\geq \sup_y g^T y - f(y) \\ g^T x - f(x) &\geq f^*(g) \end{aligned}$$

Subgradients and fenchel conjugates

Conclusion.

$$\begin{aligned}g \in \partial f(x) &\iff f^*(g) + f(x) = g^T x \\ &\iff x \in \operatorname{argmax}_x g^T x - f(x)\end{aligned}$$

consider the same implications for the function f^* :

$$\begin{aligned}x \in \partial f^*(g) &\iff f(x) + f^*(g) = x^T g \\ &\iff g \in \operatorname{argmax}_g g^T x - f^*(g)\end{aligned}$$

so all these conditions are equivalent, and $g \in \partial f(x) \iff x \in \partial f^*(g)$!

Compute subgradient via fenchel conjugate

$$\partial f(x) = \operatorname{argmax}_g g^T x - f^*(g)$$

example. let $f(x) = \|x\|_1$. compute

$$f^*(g) = \sup_x g^T x - \|x\|_1$$

=

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given x ,

$$\begin{aligned} \partial f(x) &= \operatorname{argmax}_g g^T x - f^*(g) \\ &= \operatorname{argmax}_{\|g\|_\infty \leq 1} g^T x \\ &= \mathbf{sign}(x) \end{aligned}$$

where **sign** is computed elementwise

Compute subgradient via fenchel conjugate

$$\partial f(x) = \operatorname{argmax}_g g^T x - f^*(g)$$

example. let $f(X) = \|X\|_*$. compute

$$f^*(G) = \sup_X \operatorname{tr}(G, X) - \|X\|_*$$

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where $\|G\| = \sigma_1(G)$ is the operator norm of G .

Compute subgradient via fenchel conjugate

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where $\|G\| = \sigma_1(G)$ is the operator norm of G .

given $X = U \mathbf{diag}(\sigma) V^T$,

$$\begin{aligned} \partial f(x) &= \operatorname{argmax}_G \operatorname{tr}(G, X) - f^*(G) \\ &= \operatorname{argmax}_{\|G\| \leq 1} \operatorname{tr}(G, X) \\ &= U \mathbf{diag}(\mathbf{sign}(\sigma)) V^T \end{aligned}$$

where **sign** is computed elementwise.

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Subgradient method

the **subgradient method** is a simple algorithm to minimize nondifferentiable convex function f

$$x^{(k+1)} = x^{(k)} - \alpha_k g^{(k)}$$

- ▶ $x^{(k)}$ is the k th iterate
- ▶ $g^{(k)}$ is **any** subgradient of f at $x^{(k)}$
- ▶ $\alpha_k > 0$ is the k th step size

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warning: subgradient method is **not** a descent method.

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- ▶ $g^{(k)}$ is **any** subgradient of f at $x^{(k)}$
- ▶ $\alpha_k > 0$ is the k th step size

warning: subgradient method is **not** a descent method.
instead, keep track of best point so far

$$f_{\text{best}}^{(k)} = \min_{i=1,\dots,k} f(x^{(i)})$$

How to avoid slow convergence

don't use subgradient method for very high accuracy!

instead,

- ▶ for high accuracy: rewrite problem as LP or SDP; use IPM
- ▶ for medium accuracy:

- ▶ regularize your objective (so it's strongly convex)

$$\tilde{f}(x) = f(x) + \alpha \|x - x^0\|^2$$

- ▶ smooth your objective (so it's smooth)

$$\tilde{f}(x) = \mathbb{E}_{y: \|y-x\| \leq \delta} f(y)$$

- ▶ infimal convolution (so it's smooth and strongly convex):

$$\tilde{f}(x) = \inf_y f(y) + \frac{\rho}{2} \|y - x\|^2$$

- ▶ more on these later...
 - ▶ for low accuracy: use a constant step size; terminate when you stop improving much or get bored

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define the **proximal operator** of the function $f : \mathbb{R}^d \rightarrow \mathbb{R}$

$$\mathbf{prox}_f(x) = \operatorname{argmin}_z (f(z) + \frac{1}{2}\|z - x\|_2^2)$$

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- ▶ $\mathbf{prox}_f : \mathbb{R}^d \rightarrow \mathbb{R}^d$
- ▶ **generalized projection:** if $\mathbf{1}_C$ is the indicator of set C ,

$$\mathbf{prox}_{\mathbf{1}_C}(w) = \Pi_C(w)$$

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- ▶ **implicit gradient step:** if $z = \mathbf{prox}_f(x)$

$$\begin{aligned} \partial f(z) + z - x &= 0 \\ z &= x - \partial f(z) \end{aligned}$$

Maps from functions to functions

for a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

- ▶ **prox** maps f to a new function **prox** $_f : \mathbb{R}^d \rightarrow \mathbb{R}^d$
 - ▶ **prox** $_f(x)$ evaluates this function at the point x
- ▶ ∇ maps f to a new function $\nabla f : \mathbb{R}^d \rightarrow \mathbb{R}^d$
 - ▶ $\nabla f(x)$ evaluates this function at the point x

Let's evaluate some proximal operators!

define the **proximal operator** of the function $f : \mathbb{R}^d \rightarrow \mathbb{R}$

$$\mathbf{prox}_f(x) = \operatorname{argmin}_z \left(f(z) + \frac{1}{2} \|z - x\|_2^2 \right)$$

► $f(x) = 0$

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- ▶ $f(x) = 0$ (identity)

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- ▶ $f(x) = |x|$ (soft-thresholding)

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- ▶ $f(x) = x^2$ (shrinkage)
- ▶ $f(x) = |x|$ (soft-thresholding)
- ▶ $f(x) = \mathbf{1}(x \geq 0)$

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- ▶ $f(x) = |x|$ (soft-thresholding)
- ▶ $f(x) = \mathbf{1}(x \geq 0)$ (projection)

Let's evaluate some proximal operators!

define the **proximal operator** of the function $f : \mathbb{R}^d \rightarrow \mathbb{R}$

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- ▶ $f(x) = \|x\|_1$ (soft-thresholding on each index)
- ▶ $f(X) = \|X\|_*$ (soft-thresholding on singular values)

Proxable functions

we say a function f is **proxable** if it's easy to evaluate $\text{prox}_f(x)$

all examples from previous slide are proxable

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Subgradient properties

Subgradient method

Proximal operators

Proximal gradient method

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Proximal gradient method

suppose f is smooth, g is non-smooth. solve

$$\text{minimize } f(x) + g(x)$$

using proximal operators together with gradient steps?

Proximal gradient method

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$$\text{minimize } f(x) + g(x)$$

using proximal operators together with gradient steps? idea:

$$x^+ = \mathbf{prox}_{tg}(x - t\nabla f(x))$$

- ▶ the proximal operator steps towards the minimum of g
- ▶ gradient method steps towards minimum of f

Proximal gradient: examples

with smooth loss $f(x) = \frac{1}{2}\|Ax - b\|_2^2$, regularize with

- ▶ projected gradient: $g(x) = \mathbf{1}_\Omega(x)$
- ▶ nonnegative least squares: $g(x) = \mathbf{1}_+(x)$
- ▶ lasso: $g(x) = \lambda\|x\|_1$
- ▶ ...

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Functions

in much of what follows, we'll need to assume functions are

- ▶ closed: **epi**(f) is a closed set
- ▶ convex: f is convex
- ▶ proper: **dom** f is non-empty

which we abbreviate as CCP

Relations

$(x, \partial f(x))$ and $(x, \mathbf{prox}_f(x))$ define **relations** on $\mathbb{R}^n$

- ▶ a **relation** R on $\mathbb{R}^n$ is a subset of $\mathbb{R}^n \times \mathbb{R}^n$
- ▶ $\mathbf{dom} R = \{x : (x, y) \in R\}$
- ▶ let $R(x) = \{y : (x, y) \in R\}$
- ▶ if $R(x)$ is always empty or a singleton, we say R is a function
- ▶ any function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defines a relation $\{(x, f(x)) : x \in \mathbf{dom} f\}$

Relations: examples

- ▶ empty relation: $\emptyset$
- ▶ full relation: $\mathbb{R}^n \times \mathbb{R}^n$
- ▶ identity: $\{(x, x) : x \in \mathbb{R}^n\}$
- ▶ zero: $\{(x, 0) : x \in \mathbb{R}^n\}$
- ▶ subdifferential: $\partial f = \{(x, g) : x \in \mathbf{dom} f, g \in \partial f(x)\}$

Operations on relations

if R and S are relations, define

- ▶ composition: $RS = \{(x, z) : (x, y) \in R, (y, z) \in S\}$
- ▶ addition: $R + S = \{(x, y + z) : (x, y) \in R, (x, z) \in S\}$
- ▶ inverses: $R^{-1} = \{(y, x) : (x, y) \in R\}$

use inequality on sets to mean the inequality holds for any element in the set, e.g.,

$$f(y) \geq f(x) + \partial f^T(y - x)$$

Example: fenchel conjugates and the subdifferential

recall $f^*(y) := \max_x (y^T x - f(x))$. if f is CPP, $(f^*)^* = f^{**} = f$, so

$$\begin{aligned}(u, v) \in (\partial f)^{-1} &\iff (v, u) \in \partial f \\ &\iff u \in \partial f(v) \\ &\iff 0 \in \partial f(v) - u \\ &\iff v \in \operatorname{argmin}_x (f(x) - u^T x) \\ &\iff v \in \operatorname{argmax}_x (u^T x - f(x)) \\ &\iff f(v) + f^*(u) = u^T v \\ &\iff u \in \operatorname{argmax}_y (y^T v - f^*(y)) \\ &\iff 0 \in v - \partial f^*(u) \\ &\iff (u, v) \in \partial f^*\end{aligned}$$

this shows $\partial f^* = \partial f^{-1}$

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Zeros of a relation

- ▶ x is a **zero** of R if $0 \in R(x)$
- ▶ the **zero set** of R is $R^{-1}(0) = \{x : (x, 0) \in R\}$

Zeros of a relation

- ▶ x is a **zero** of R if $0 \in R(x)$
- ▶ the **zero set** of R is $R^{-1}(0) = \{x : (x, 0) \in R\}$

x is a zero of ∂f iff x solves minimize $f(x)$

Lipschitz operators

relation F has Lipschitz constant L if for all $(x, u) \in F$ and $(y, v) \in F$,

$$\|u - v\| \leq L\|x - y\|$$

fact: if F is Lipschitz, then F is a function.

proof:

Lipschitz operators

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fact: if F is Lipschitz, then F is a function.

proof: if $(x, u) \in F$ and $(x, v) \in F$,

$$\|u - v\| \leq L\|x - x\| = 0$$

- ▶ the relation F is **nonexpansive** if $L \leq 1$
- ▶ the relation F is **contractive** if $L < 1$

examples: rotation and translation are nonexpansive; scaling by $c < 1$ is contractive.

Gradient update is contractive for SSC functions

suppose f is α -strongly convex and β -smooth. the relation

$$I - t\nabla f = \{(x, x - t\nabla f(x)) : x \in \mathbf{dom} f\}$$

is Lipschitz with parameter $L = \max\{|1 - t\alpha|, |1 - t\beta|\}$.

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 $L = \frac{\kappa - 1}{\kappa + 1}$

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hint: use the fundamental theorem of calculus

$$(I - t\nabla f)(x) - (I - t\nabla f)(y) = \int_0^1 (I - t\nabla^2 f(\theta x + (1 - \theta)y))(x - y) d\theta$$

and Jensen's inequality

$$\left\| \int_0^1 v(t) dt \right\| \leq \int_0^1 \|v(t)\| dt$$

source: Ryu and Yin (2022)

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is Lipschitz with parameter $L = \max\{|1 - t\alpha|, |1 - t\beta|\}$.

proof:

$$\begin{aligned} & \| (I - t\nabla f)(x) - (I - t\nabla f)(y) \| \\ &= \left\| \int_0^1 (I - t\nabla^2 f(\theta x + (1 - \theta)y))(x - y) d\theta \right\| \\ &\leq \int_0^1 \| (I - t\nabla^2 f(\theta x + (1 - \theta)y))(x - y) \| d\theta \\ &\leq \int_0^1 \max(|1 - t\alpha|, |1 - t\beta|) d\theta \|x - y\| \\ &= \max(|1 - t\alpha|, |1 - t\beta|) \|x - y\| \end{aligned}$$

last ineq uses $\alpha I \preceq \nabla^2 f \preceq \beta I \implies (1 - t\beta)I \preceq I - t\nabla^2 f \preceq (1 - t\alpha)I$

Proximal map is nonexpansive

the proximal map of any convex function f is nonexpansive:

$$\|\mathbf{prox}_f(y) - \mathbf{prox}_f(x)\| \leq \|y - x\|$$

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proof: let $u = \mathbf{prox}_f(x)$ and $v = \mathbf{prox}_f(y)$, so

$$x - u \in \partial f(u), \quad y - v \in \partial f(v)$$

then by the subgradient inequality,

$$f(v) \geq f(u) + \langle x - u, v - u \rangle \quad \text{and} \quad f(u) \geq f(v) + \langle y - v, u - v \rangle$$

add these to show

$$\begin{aligned} 0 &\geq \langle y - x + u - v, u - v \rangle \\ \langle x - y, u - v \rangle &\geq \|u - v\|^2 \\ \|x - y\| &\geq \|u - v\| \end{aligned}$$

- ▶ second line shows $\mathbf{prox}_f$ is **firmly nonexpansive**
- ▶ third line uses Cauchy-Schwarz to show it is nonexpansive

Proximal map is contractive for SC functions

the proximal map of an α -SC function f is $\frac{1}{1+2\alpha}$ -contractive:

$$\|\mathbf{prox}_f(y) - \mathbf{prox}_f(x)\| \leq \frac{1}{1+2\alpha} \|y - x\|$$

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proof: let $u = \mathbf{prox}_f(x)$ and $v = \mathbf{prox}_f(y)$, so

$$x - u \in \partial f(u), \quad y - v \in \partial f(v)$$

by strong convexity

$$f(v) \geq f(u) + \langle x - u, v - u \rangle + \alpha \|v - u\|^2$$

$$f(u) \geq f(v) + \langle y - v, u - v \rangle + \alpha \|u - v\|^2$$

add these to show

$$0 \geq \langle y - x + u - v, u - v \rangle + 2\alpha \|u - v\|^2$$

$$\langle x - y, u - v \rangle \geq (1 + 2\alpha) \|u - v\|^2$$

$$\frac{1}{1+2\alpha} \|x - y\| \geq \|u - v\|$$

Fixed points

x is a **fixed point** of F if $x = F(x)$

examples:

- ▶ $F(x) = x$: every point is a fixed point
- ▶ $F(x) = 0$: only 0 is a fixed point

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- ▶ a nonexpansive operator F need not have a fixed point
proof: translation

Fixed point of proximal gradient map

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is proxable, and let $t > 0$.

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Proof. Use the definition of the proximal operator to check

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The point x minimizes $f + g$ iff

$$\begin{aligned} 0 &\in \nabla f(x) + \partial g(x) \\ x &\in t\nabla f(x) + x + t\partial g(x) \\ (I - t\nabla f)(x) &\in (I + t\partial g)(x) \\ x &\in (I + t\partial g)^{-1}(I - t\nabla f)(x) \end{aligned}$$

Fixed point iteration

to find a fixed point of F , try the fixed point iteration

$$x^{(k+1)} = F(x^{(k)})$$

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Q: when does this converge?

Fixed point iteration: contractive

Banach fixed point theorem: if F is a contraction, the iteration

$$x^{(k+1)} = F(x^{(k)})$$

converges to the unique fixed point of F

properties: if L is the Lipschitz constant of F ,

- ▶ distance to fixed point decreases monotonically:

$$\|x^{(k+1)} - x^*\| = \|F(x^{(k)}) - F(x^*)\| \leq L\|x^{(k)} - x^*\|$$

(iteration is **Fejer-monotone**)

- ▶ linear convergence with rate L

Proof

proof:

Proof

proof: if F has Lipschitz constant $L < 1$,

► sequence $x^{(k)}$ is Cauchy:

$$\begin{aligned}\|x^{(k+\ell)} - x^{(k)}\| &\leq \|x^{(k+\ell)} - x^{(k+\ell-1)}\| + \dots + \|x^{(k+1)} - x^{(k)}\| \\ &\leq (L^{\ell-1} + \dots + 1)\|x^{(k+1)} - x^{(k)}\| \\ &\leq \frac{1}{1-L}\|x^{(k+1)} - x^{(k)}\| \\ &\leq \frac{L^k}{1-L}\|x^{(1)} - x^{(0)}\|\end{aligned}$$

► so it converges to a point x^* . must be the (unique) FP!

► converges to x^* linearly with rate L

$$\|x^{(k)} - x^*\| = \|F(x^{(k-1)}) - F(x^*)\| \leq L\|x^{(k-1)} - x^*\| \leq L^k\|x^{(0)} - x^*\|$$

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Fixed point iteration: nonexpansive

if F is nonexpansive, the iteration

$$x^{(k+1)} = F(x^{(k)})$$

need not converge to a fixed point even if one exists.

proof:

Fixed point iteration: nonexpansive

if F is nonexpansive, the iteration

$$x^{(k+1)} = F(x^{(k)})$$

need not converge to a fixed point even if one exists.

proof:

- ▶ let F rotate its argument by θ degrees around the origin.
- ▶ then F is nonexpansive and has a fixed point at $x^* = 0$.
- ▶ but if $\|x^{(0)}\| = r$, then $\|F(x^{(k)})\| = r$ for all k .

Averaged operators

an operator F is **averaged** if

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for $\theta \in (0, 1)$, G nonexpansive

fact: if F is averaged, then x is FP of $F \iff x$ is FP of G

proof:

$$\begin{aligned}x &= Fx = \theta Gx + (1 - \theta)Ix = \theta Gx + (1 - \theta)x \\ \theta x &= \theta Gx \\ x &= Gx\end{aligned}$$

$\implies$ if G is nonexpansive, $F = \frac{1}{2}I + \frac{1}{2}G$ is averaged with same FPs

Fixed point iteration: averaged

if $F = \theta G + (1 - \theta)I$ is averaged ($\theta \in (0, 1)$, G nonexpansive),
the iteration

$$x^{(k+1)} = F(x^{(k)})$$

converges to a fixed point if one exists.

(also called the damped, averaged, or Mann-Krasnosel'skii iteration.)

properties: Ryu and Boyd (2016)

- ▶ distance to fixed point decreases monotonically (Fejer-monotone)
- ▶ sublinear convergence of fixed point residual

$$\|Gx^{(k)} - x^{(k)}\|^2 \leq \frac{1}{(k+1)\theta(1-\theta)} \|x^{(0)} - x^*\|^2$$

Gradient descent operator is averaged

follows Ryu and Yin (2022)

fact: if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is β -smooth, then $I - \frac{2}{\beta} \nabla f$ is non-expansive

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follows Ryu and Yin (2022)

fact: if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is β -smooth, then $I - \frac{2}{\beta} \nabla f$ is non-expansive

proof: since f is β -smooth,

$$\begin{aligned} \|(I - \frac{2}{\beta} \nabla f)(x) - (I - \frac{2}{\beta} \nabla f)(y)\|^2 &= \|x - y\|^2 - \frac{4}{\beta} \langle x - y, \nabla f(x) - \nabla f(y) \rangle \\ &\quad + \frac{4}{\beta^2} \|\nabla f(x) - \nabla f(y)\|^2 \\ &\leq \|x - y\|^2 \end{aligned}$$

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corollary: if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is β -smooth, then $I - t \nabla f$ is averaged for $t \in (0, \frac{2}{\beta})$

since $I - t \nabla f = (1 - \frac{t\beta}{2})I + \frac{t\beta}{2}(I - \frac{2}{\beta} \nabla f)$

When does proximal gradient converge?

proximal gradient converges at rate $O(1/k)$ when $I - t\nabla f$ is averaged and $\mathbf{prox}_{tg}$ is nonexpansive

- ▶ if f is β -smooth and step size $t \in (0, \frac{2}{\beta})$
- ▶ and g is convex

proximal gradient converges linearly when, in addition, $I - t\nabla f$ or $\mathbf{prox}_{tg}$ is contractive

- ▶ if f is β -smooth and α -strongly convex and $\max(|1 - t\alpha|, |1 - t\beta|) < 1$
- ▶ or if g is strongly convex

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Q: How fast does proximal gradient converge for the lasso? for elastic net? for bounded least squares? for bounded least squares with an ℓ_2 regularizer? for ℓ_2 -regularized logistic regression?

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fixed point iteration using prox is called **proximal point method**

$$x^{(k+1)} = \mathbf{prox}_{tf}(x^{(k)})$$

properties:

- ▶ $\mathbf{prox}_{tf}$ is $\frac{1}{2}$ averaged for any $t > 0$, so
- ▶ converges for any $t > 0$
- ▶ to a zero of ∂f (= FPs of $\mathbf{prox}_{tf}$)
- ▶ if f is strongly convex, $\mathbf{prox}_{tf}$ is a contraction, so converges linearly
- ▶ not usually a practical method (often, as hard as solving original problem)

Method of multipliers

consider

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & Ax = b\end{array}$$

let

$$g(\mu) = -(\inf_x f(x) + \lambda^T (Ax - b)) = f^*(-A^T \lambda) + \lambda^T b$$

be the (negative) dual function, and consider the proximal point method for $t > 0$

$$\lambda^{(k+1)} = \mathbf{prox}_{tg}(\lambda^{(k)})$$

- ▶ $\partial g(\lambda) = -A\partial(f^*(-A^T \lambda)) + b$
- ▶ $x \in \partial(f^*(-A^T \lambda))$ iff $-A^T \lambda \in \partial f(x)$
- ▶ so if $\lambda' = \mathbf{prox}_{tg}(\lambda) = (I + t\partial g)^{-1}(\lambda)$, then

$$\lambda \in \lambda' + t\partial g(\lambda')$$

$$\lambda = \lambda' - t(Ax - b) \quad \text{for some } x \text{ with } -A^T \lambda' \in \partial f(x)$$

Method of multipliers

notice x minimizes the **Augmented Lagrangian** $L_t(x, y)$

$$0 \in \partial f(x) + A^T(\lambda + t(Ax - b))$$

$$x \in \operatorname{argmin}_x f(x) + \lambda^T(Ax - b) + t/2\|Ax - b\|^2 = L_t(x, \lambda)$$

so proximal point method for g is

$$x^{(k+1)} \in \operatorname{argmin}_x L_t(x, \lambda^{(k)})$$

$$\lambda^{(k+1)} = \lambda^{(k)} + t(Ax^{(k+1)} - b)$$

also called the **method of multipliers**

properties:

- ▶ always converges
- ▶ if f is smooth, then g is strongly convex, prox_{tg} is a contraction, and the method of multipliers converges linearly
- ▶ useful if f is smooth and A is very sparse